ClipSAT : Differential Equations First/Second Order, Laplace, Systems & S [ANSWER KEY]

Unit 9.01: Numerical Methods: Euler, RK2, RK4 & Stability

Q1. The second-order linear homogeneous ODE f''x - 39f'x + 350fx = 0 has characteristic roots lambda_1, lambda_2 . Determine the trace sum lambda_1 + lambda_2 .

[x] A: 39

[ ] B: 24

[ ] C: 42

[ ] D: 53

Solution: From lambda^2 - 39lambda + 350 = 0 , the sum of roots is 39 .

Q2. Consider the linear first-order differential equation frac{dy}{dx} + 2y = 6 . If the initial condition is y0 = 1 , find the value of y1 .

[ ] A: 2

[ ] B: e^2 - 1

[ ] C: 3 - 2e^{-2}

[x] D: 3 - 2e^{-1}

Solution: The given differential equation is a linear first-order ODE of the form frac{dy}{dx} + Pxy = Qx , where Px = 2 and Qx = 6 . The integrating factor mux is given by mux = e^{int Px dx} . Calculating the integrating factor: mux = e^{int 2 dx} = e^{2x} . Multiplying the differential equation by the integrating factor: e^{2x}frac{dy}{dx} + 2e^{2x}y = 6e^{2x} . The left side is the derivative of the product ymux : frac{d}{dx}ye^{2x} = 6e^{2x} . Integrating both sides with respect to x : ye^{2x} = int 6e^{2x} dx . ye^{2x} = 6 cdot frac{1}{2}e^{2x} + C . ye^{2x} = 3e^{2x} + C . Solving for y : y = 3 + Ce^{-2x} . Using the initial condition y0 = 1 to find C : 1 = 3 + Ce^{-20} . 1 = 3 + C . C = 1 - 3 = -2 . So, the particular solution is yx = 3 - 2e^{-2x} . Now, we need to find y1 : y1 = 3 - 2e^{-21} = 3 - 2e^{-2} . This matches choice C.

Q3. Consider the differential equation dfrac{dy}{dx} = x + y with the initial condition y0 = 1 . Use the Euler method with a step size of h = 0.1 to approximate y0.2 .

[ ] A: 1.200

[ ] B: 1.210

[ ] C: 1.230

[x] D: 1.220

Solution: Using Euler's method y_{n+1} = y_n + hx_n + y_n with x_0 = 0, y_0 = 1, h = 0.1 : 1. y_1 = 1 + 0.10 + 1 = 1.100 at x_1 = 0.1 . 2. y_2 = 1.1 + 0.10.1 + 1.1 = 1.1 + 0.12 = 1.220 at x_2 = 0.2 .

Q4. Consider the second-order linear homogeneous differential equation y'' - 5y' + 6y = 0 . If the auxiliary equation has roots r_1 and r_2 , find the value of r_1 + r_2 .

[x] A: 5

[ ] B: -5

[ ] C: 6

[ ] D: -6

Solution: The given differential equation is y'' - 5y' + 6y = 0 . The auxiliary equation for a second-order linear homogeneous differential equation of the form ay'' + by' + cy = 0 is ar^2 + br + c = 0 . In this case, a=1 , b=-5 , and c=6 . So the auxiliary equation is r^2 - 5r + 6 = 0 . Let the roots of this quadratic equation be r_1 and r_2 . According to Vieta's formulas, for a quadratic equation ar^2 + br + c = 0 , the sum of the roots r_1 + r_2 is equal to -frac{b}{a} . In our auxiliary equation r^2 - 5r + 6 = 0 , a=1 and b=-5 . Therefore, the sum of the roots is r_1 + r_2 = - frac{-5}{1} = 5 .

Q5. Consider the separable differential equation dfrac{dy}{dx} = 2x with the initial condition y0 = 3 . Find the value of y2 .

[x] A: 7
[ ] B: 6
[ ] C: 8
[ ] D: 5

Solution: The differential equation is dfrac{dy}{dx} = 2x . This is a separable equation. We can rewrite it as dy = 2x , dx . Integrating both sides gives int dy = int 2x , dx , which results in y = x^2 + C , where C is the constant of integration. We are given the initial condition y0 = 3 . Substituting these values into the general solution, we get 3 = 0^2 + C , so C = 3 . Therefore, the particular solution is y = x^2 + 3 . We need to find the value of y2 . Substituting x = 2 into the particular solution, we have y2 = 2^2 + 3 = 4 + 3 = 7 .

Q6. Consider the differential equation y'' - xy = 0 . The point x=0 is an ordinary point. If we assume a power series solution of the form yx = sum_{n=0}^{infty} c_n x^n , find the recurrence relation for the coefficients c_n .

[x] A: c_{n+2} = frac{c_{n-1}}{ n+2n+1 } fo
[ ] B: c_{n+2} = frac{c_n}{ n+2n+1 } for n
[ ] C: c_{n+2} = -frac{c_{n-1}}{ n+2n+1 } f
[ ] D: c_{n+2} = -frac{c_n}{ n+2n+1 } for n

Solution: We are given the differential equation y'' - xy = 0 and assume a power series solution of the form yx = sum_{n=0}^{infty} c_n x^n . newline Differentiating twice with respect to x , we get: y'x = sum_{n=1}^{infty} n c_n x^{n-1} and y''x = sum_{n=2}^{infty} nn-1 c_n x^{n-2} . newline Substituting these into the differential equation: sum_{n=2}^{infty} nn-1 c_n x^{n-2} - x sum_{n=0}^{infty} c_n x^n = 0 . newline We can rewrite the second term as sum_{n=0}^{infty} c_n x^{n+1} . To combine the series, we need the powers of x to be the same. Let k = n-2 in the first series, so n = k+2 . When n=2, k=0 . The first series becomes sum_{k=0}^{infty} k+2k+1 c_{k+2} x^k . newline Let k = n+1 in the second series, so n = k-1 . When n=0, k=1 . The second series becomes sum_{k=1}^{infty} c_{k-1} x^k . newline So the equation is sum_{k=0}^{infty} k+2k+1 c_{k+2} x^k - sum_{k=1}^{infty} c_{k-1} x^k = 0 . newline We can separate the k=0 term from the first sum: 21c_2 + sum_{k=1}^{infty} k+2k+1 c_{k+2} x^k - sum_{k=1}^{infty} c_{k-1} x^k = 0 . newline For this equation to hold for all x , the coefficients of each power of x must be zero. newline For k=0 : 2c_2 = 0 implies c_2 = 0 . newline For k geq 1 : k+2k+1 c_{k+2} - c_{k-1} = 0 . newline Thus, the recurrence relation is c_{k+2} = frac{c_{k-1}}{k+2k+1} for k geq 1 . Replacing k with n , we get c_{n+2} = frac{c_{n-1}}{n+2n+1} for n geq 1 .

Q7. The second-order linear homogeneous ODE u''s - 40u's + 144us = 0 has characteristic roots lambda_1, lambda_2 . Determine the trace sum lambda_1 + lambda_2 .

[ ] A: 44
[ ] B: 43
[ ] C: 24
[x] D: 40

Solution: From lambda^2 - 40lambda + 144 = 0 , the sum of roots is 40 .

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Q8. The Laplace transform of us = 170e^{19s} is vs = dfrac{170}{s - 19} for s > 19 . Determine the evaluated transform v20 .

[x] A: 170

[ ] B: 151

[ ] C: 173

[ ] D: 187

Solution: Substituting s = 20 gives frac{170}{20 - 19} = 170 .